



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Tableau Desktop Foundations

TGS Ref No: TGS-2025053175

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 5

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
5	5 September 2026	Retitled to Tableau Desktop Foundations; added Tableau Prep Builder data cleansing to Topic 1 and new Lab 11.	Dr. Alfred Ang

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Course Information

Course Title	Tableau Desktop Foundations
Course Code	TGS-2025053175
TSC	Data Analytics · ATP-PIN-2001-1.1
Duration	2 days · 16 training hours, including 2-hour final assessment
Daily Time	9:00 am–6:00 pm; 1-hour lunch; two 10-minute tea breaks
Mode	Instructor-led concept teaching, demonstrations and Tableau Desktop labs
Trainer	Dr. Alfred Ang

Learning Outcomes

- LO1: Identify and connect appropriate data sources, then prepare trustworthy Tableau data models.
- LO2: Organise fields and create effective visual elements that answer business questions.
- LO3: Extract, combine and analyse data using relationships, joins, unions, filters and calculations.
- LO4: Build, format and share interactive dashboards and workbooks that communicate meaningful insights.

Assessment

- Written Assessment (WA/SAQ) — 4 open-ended questions, 1 hour, open book.
- Practical Performance (PP) — 4 Tableau tasks, 1 hour, open book.
- Assessment runs 4:00–6:00 pm on Day 2.
- Open book: course slides, Learner Guide and approved materials only.
- Learners need at least 75% attendance and a Competent assessment outcome for funding eligibility.

Daily Schedule

Day 1 — Connecting, preparing, exploring and analysing data

Time	Duration	Topic / Lab	Method	Slides
9:00–9:30	30 min	Welcome, attendance, trainer/learner introductions, outcomes and exam orientation	Facilitation	Slides 1–18
9:30–10:30	60 min	Topic 1 — Connecting to and Preparing Data, including Tableau Prep Builder cleansing	Concept + demo	Slides 19–63

10:30–10:40	10 min	Tea break	Break	—
10:40–12:40	120 min	Labs 1–3 — connection/extract audit, relationship model, union and join	Guided hands-on	Slides 55–60
12:40–13:00	20 min	Lab 11 — Tableau Prep cleansing flow: profile, clean, split and Group and Replace	Guided hands-on	Slides 61–62
13:00–14:00	60 min	Lunch break	Lunch	—
14:00–14:55	55 min	Lab 11 continued — pivot, union, join, output the flow and validate against Control	Guided hands-on	Slides 61–62
14:55–15:45	50 min	Topic 2 — Exploring and Analyzing Data	Concept + demo	Slides 64–103
15:45–15:55	10 min	Tea break	Break	—
15:55–17:40	105 min	Labs 4–7 — chart portfolio, organisation/filtering, calculations and geographic analysis	Guided hands-on	Slides 95–102
17:40–18:00	20 min	Day 1 exam check, Q&A and PM digital attendance	Review	Slides 64–103

Total training time excluding lunch: 480 minutes (8 hours).

Day 2 — Dashboards, sharing, core concepts and certification readiness

Time	Duration	Topic / Lab	Method	Slides
9:00–9:15	15 min	Day 1 recap and AM digital attendance	Review	Slides 64–103
9:15–10:30	75 min	Topic 3 — Sharing Insights with Dashboards and Workbooks	Concept + demo	Slides 104–132
10:30–10:40	10 min	Tea break	Break	—
10:40–12:30	110 min	Labs 8–9 — interactive dashboard and governed executive sharing	Guided hands-on	Slides 128–131
12:30–13:00	30 min	Topic 4 concepts and certification reasoning	Concept + demo	Slides 133–153
13:00–14:00	60 min	Lunch break	Lunch	—
14:00–15:40	100 min	Lab 10 — Northstar Retail capstone and exam-domain evidence	Guided hands-on	Slides 151–152
15:40–15:50	10 min	Tea break	Break	—
15:50–16:00	10 min	Briefing for Assessment	Assessment	Slides 16

16:00–17:00	60 min	Written Assessment (WA/SAQ) — 4 open-ended questions, open book	Assessment	—
17:00–18:00	60 min	Practical Performance (PP) — 4 Tableau tasks, open book; Assessment attendance	Assessment	—

Total training time excluding lunch: 480 minutes (8 hours).

Topic and Lab Slide Map

Topic 01 — Connecting to and Preparing Data in Tableau Desktop • Slides 19–63

Official weighting: 23%. Connections · extracts · metadata · relationships · joins · unions · field properties · Tableau Prep cleansing

Lab 01 — Audit a Retail Sales Connection and Extract • Slides 55–56

Goal: Connect to Excel, validate field metadata, compare live and extract behaviour.

Duration: 60 minutes. Evidence: A validated Tableau data source, field-role audit, live/extract decision and refreshed .hyper extract

Lab 02 — Model Orders, Returns, Products and Targets • Slides 57–58

Goal: Create relationships and compare them with physical joins. Duration: 75 minutes.

Evidence: A four-table logical model with validated cardinality, unmatched-key checks and a join-risk comparison

Lab 03 — Union Monthly Files and Join Store Attributes • Slides 59–60

Goal: Append compatible tables, merge mismatched fields and join a lookup. Duration: 60 minutes. Evidence: A three-month union with provenance fields, merged columns, a validated store join and saved data-source metadata

Lab 11 — Cleanse a Messy Sales Extract with Tableau Prep Builder • Slides 61–62

Goal: Build a repeatable Tableau Prep flow that profiles, cleans, reshapes and outputs an analysis-ready table. Duration: 75 minutes. Evidence: A saved Tableau Prep flow (.tfl) with Input, Clean, Split, Group and Replace, Pivot, Union, Join and Output steps, plus a cleaned .hyper or CSV output and a documented change list

Topic 02 — Exploring and Analyzing Data in Tableau Desktop • Slides 64–103

Official weighting: 37%. Charts · groups · sets · hierarchies · filters · analytics · calculations

Lab 04 — Build the Core Tableau Chart Portfolio • Slides 95–96

Goal: Create and justify nine chart patterns from one realistic sales dataset. Duration: 75 minutes. Evidence: Nine named worksheets: bar, line, scatterplot, symbol map, combined axis, dual axis, stacked bar, density map and highlight table

Lab 05 — Create Groups, Sets, Hierarchies and Filters • Slides 97–98

Goal: Organise customer data and apply filters at the intended analytical stage. Duration: 60 minutes. Evidence: Customer tier groups, Top-N and threshold sets, drill hierarchy, context filter and relative date controls

Lab 06 — Apply Calculations, Table Calculations and Parameters • Slides 99–100

Goal: Create reliable row-level, aggregate and view-dependent analytics. Duration: 75 minutes. Evidence: Calculated KPIs, running totals, percent-of-total, moving average, reference line and parameter-driven measure swap

Lab 07 — Analyse Southeast Asia Logistics Geographically • Slides 101–102

Goal: Create symbol, filled and density maps with trustworthy geographic context. Duration: 60 minutes. Evidence: Resolved geographic roles, three map types, tooltip context and a map-vs-bar decision note

Topic 03 — Sharing Insights with Dashboards and Workbooks • Slides 104–132

Official weighting: 25%. Formatting · dashboards · actions · device layouts · stories · export · publish

Lab 08 — Build an Interactive Service Operations Dashboard • Slides 128–129

Goal: Compose worksheets, containers, filters, tooltips and dashboard actions. Duration: 90 minutes. Evidence: A 1366×768 operational dashboard with KPIs, four views, scoped filters, highlight/filter actions and accessible tooltips

Lab 09 — Optimise, Package and Share an Executive Workbook • Slides 130–131

Goal: Create device layouts, a story, exports and a governed sharing plan. Duration: 60 minutes. Evidence: Desktop and phone layouts, three story points, TWB/TWBX comparison, PDF/PNG/PPT exports and a publishing checklist

Topic 04 — Understanding Tableau Concepts and Certification Preparation • Slides 133–153

Official weighting: 15%. Dimensions · measures · discrete · continuous · aggregation · exam strategy

Lab 10 — Capstone: Northstar Retail Executive Workbook • Slides 151–152

Goal: Integrate all four exam domains and complete certification-readiness evidence. Duration: 100 minutes. Evidence: A validated multi-table model, eight analytical worksheets, executive dashboard, phone layout, story, packaged workbook and exam-domain checklist

Resources Required

- Tableau Desktop (recommended version matching the certification guide where available)
- Microsoft Excel or compatible viewer
- Ten individual lab folders with Excel workbooks, starter.twb and acceptance checklists
- Learner Guide and course slides
- LMS/TMS access for courseware, attendance and assessment